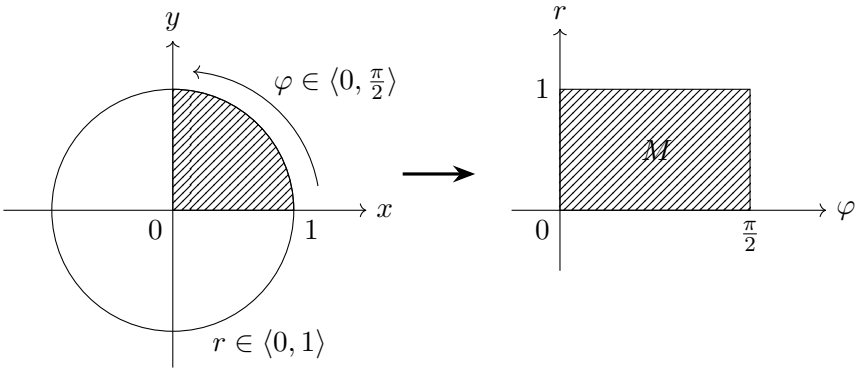
Vypočti pomocí transformace do polárních souřadnic

$$\iint_A (2+3x)\,\mathrm{d}x\,\mathrm{d}y \qquad \text{kde } A=\{[x,y]:\ \underbrace{x^2+y^2\le 1}_{\substack{\text{hraniční křivka:}\\ \text{kružnice } S[0,0],\ r=1}}\ ,\ \underbrace{x\ge 0,\ y\ge 0}_{\substack{\text{1. kvadrant}\\ \Downarrow\\ \varphi\in\langle 0{,}90^\circ\rangle=\langle 0,\frac{\pi}{2}\rangle}}\}$$



$$\iint_M (2+3(r\cos\varphi))\ \underbrace{r}_{\substack{\text{VŽDY}\\ \text{připisuji!}}}\ \mathrm{d}M=\int_0^{\frac{\pi}{2}}\left(\int_0^1\left(2r+3r^2\cos\varphi\right)\mathrm{d}r\right)\mathrm{d}\varphi=$$

$$=\int_0^{\frac{\pi}{2}}\left[\cancel{2}\,\frac{r^2}{\cancel{2}}+\cancel{3}\cos\varphi\,\frac{r^3}{\cancel{3}}\right]_0^1\mathrm{d}\varphi=\int_0^{\frac{\pi}{2}}(1+1\cdot\cos\varphi)\,\mathrm{d}\varphi=$$

$$=\Big[\varphi+\sin\varphi\Big]_0^{\frac{\pi}{2}}=\frac{\pi}{2}+\underbrace{\sin\frac{\pi}{2}}_{=1}-(0+\sin 0)=\frac{\pi}{2}+1-0=\boxed{\frac{\pi}{2}+1}$$